

Clinic IT — Packet Tracer Instructions & IEEE References

Below are step-by-step instructions to recreate the designed network in Cisco Packet Tracer, plus configuration snippets you can paste into device CLIs.

Step-by-step Packet Tracer build (high-level):

1. Place one Router (e.g., 2811) and one Layer-3 Switch (e.g., 3560) on the workspace.
2. Add three Access Switches (2960) for Emergency, Outpatient, Radiology and connect them to the Core Switch via Gigabit links.
3. Add PCs for each department (3-6 per dept depending on load) and connect to Access Switch ports.
4. Add Server devices: Database Server, Imaging Server, Backup Server and connect them to the Core Switch.
5. Configure VLANs on the Core Switch (10,20,30,40) and assign IP interfaces as default gateways.
6. Configure trunk links between Core Switch and Access Switches (dot1q).
7. Configure DHCP pools on the Core Router or dedicated DHCP server.
8. Configure NAT on the Core Router if internet access from private networks is required.
9. Implement ACLs on the Core Router to restrict inter-VLAN traffic as needed.
10. Test connectivity and file transfer speeds, and simulate radiology image transfers.

Configuration Snippets (paste these into each device CLI in Packet Tracer):

Core Router:

```
! Core Router Configuration (paste into Packet Tracer Router CLI)
hostname CoreRouter
enable secret cisco123
no ip domain-lookup

interface GigabitEthernet0/0
description Link to ISP
ip address dhcp
no shutdown

interface GigabitEthernet0/1
description Connection to Core Switch
ip address 10.0.0.1 255.255.255.0
no shutdown

ip route 0.0.0.0 0.0.0.0 GigabitEthernet0/0

! NAT example (if using internal private addressing)
access-list 1 permit 10.0.0.0 0.0.0.255
ip nat inside source list 1 interface GigabitEthernet0/0 overload

line vty 0 4
password cisco
login
!
```

Core Switch (Layer 3):

! Core Layer-3 Switch Configuration (for Packet Tracer)

hostname CoreSwitch

no ip domain-lookup

ip routing

interface Vlan10

description Emergency VLAN

ip address 10.10.10.1 255.255.255.0

interface Vlan20

description Outpatient VLAN

ip address 10.10.20.1 255.255.255.0

interface Vlan30

description Radiology VLAN

ip address 10.10.30.1 255.255.255.0

interface Vlan40

description Administration VLAN

ip address 10.10.40.1 255.255.255.0

! Trunk to access switches (example)

interface GigabitEthernet1/1

switchport trunk encapsulation dot1q

switchport mode trunk

no shutdown

! Static route to internet via CoreRouter

ip route 0.0.0.0 0.0.0.0 10.0.0.1

Access Switch (example):

! Access Switch Configuration (Departmental Switch)

hostname AccessSwitch_ED

no ip domain-lookup

vlan 10

name Emergency

vlan 20

name Outpatient

vlan 30

name Radiology

vlan 40

name Administration

interface range GigabitEthernet0/1 - 24

switchport mode access

```
switchport access vlan 10
spanning-tree portfast
no shutdown
```

! Modify interface assignments per department ports in Packet Tracer GUI

DHCP configuration (on router or server):

! DHCP Pool configuration on CoreSwitch (or Router)

```
ip dhcp excluded-address 10.10.10.1 10.10.10.10
ip dhcp excluded-address 10.10.20.1 10.10.20.10
ip dhcp excluded-address 10.10.30.1 10.10.30.10
ip dhcp excluded-address 10.10.40.1 10.10.40.10
```

```
ip dhcp pool EMERGENCY
network 10.10.10.0 255.255.255.0
default-router 10.10.10.1
dns-server 8.8.8.8
```

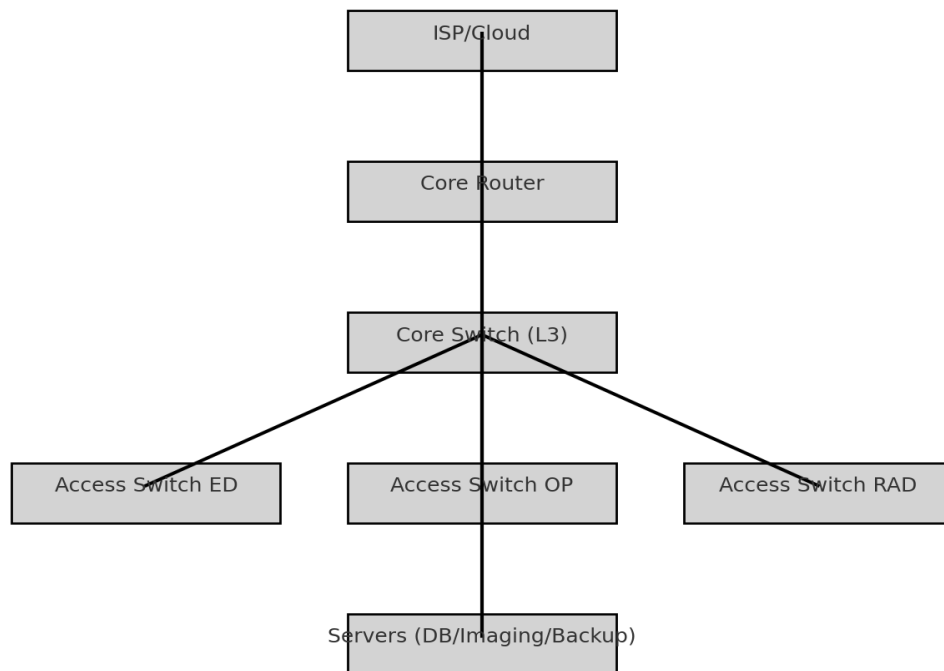
```
ip dhcp pool OUTPATIENT
network 10.10.20.0 255.255.255.0
default-router 10.10.20.1
dns-server 8.8.8.8
```

```
ip dhcp pool RADIOLOGY
network 10.10.30.0 255.255.255.0
default-router 10.10.30.1
dns-server 8.8.8.8
```

```
ip dhcp pool ADMIN
network 10.10.40.0 255.255.255.0
default-router 10.10.40.1
dns-server 8.8.8.8
```

Topology Diagram (high-level):

Clinic Network Topology (high-level)



IEEE References

- [1] Cisco Systems, "VLANs and Trunking Configuration Guide," Cisco, 2024. [Online]. Available: <https://www.cisco.com/>
- [2] Cisco Systems, "Configuring DHCP on Cisco IOS," Cisco Documentation, 2023. [Online]. Available: <https://www.cisco.com/>
- [3] Novabench, "Novabench Benchmarking Software," Novabench, 2024. [Online]. Available: <https://www.novabench.com/>
- [4] IEEE, "Cite Them Right: IEEE referencing guide," 2024. [Online]. Available: <https://www.citethemightutorial.com/>
- [5] NVIDIA, "CUDA Toolkit Documentation," NVIDIA, 2024. [Online]. Available: <https://developer.nvidia.com/cuda-toolkit>
- [6] Hennessy, J. L., & Patterson, D. A., Computer Architecture: A Quantitative Approach, 6th ed., Morgan Kaufmann, 2017.